In [10]:

```python
#Name : Harshal Veer
#Roll no :75
import numpy as np

# ART1 Algorithm (Simplified)
class ART1:
    def __init__(self, n_features, vigilance=0.5):
        self.n_features = n_features
        self.vigilance = vigilance
        self.weights = []    # cluster weights

    def _similarity(self, x, w):
        return np.sum(np.minimum(x, w)) / (np.sum(x) + 1e-9)

    def fit(self, X):
        for x in X:
            assigned = False

            for i, w in enumerate(self.weights):
                sim = self._similarity(x, w)

                if sim >= self.vigilance:
                    # Update weights
                    self.weights[i] = np.minimum(w, x)
                    assigned = True
                    break

            if not assigned:
                # Create new cluster
                self.weights.append(x.copy())

    def predict(self, X):
        labels = []
        for x in X:
            sims = [self._similarity(x, w) for w in self.weights]
            labels.append(np.argmax(sims))
        return labels


#SampleBinary Data (ART1 requires binary input)
X = np.array([
[1,1,0,0],
[1,0,0,0],
[0,1,0,0],
[0,0,1,1],
[0,0,1,0],
[0,0,0,1]
])

# Create ART model
art = ART1(n_features=4, vigilance=0.6)

#Trainmodel
art.fit(X)
```

```python

#Predict clusters
labels = art.predict(X)

# Output
print("Input Data:\n", X)
print("Cluster Labels:", labels)
print("Number of Clusters:", len(art.weights))
```

```
Input Data:
 [[1 1 0 0]
 [1  0  0  0]
 [0  1  0  0]
 [0  0  1  1]
 [0  0  1  0]
 [0 0 0 1]]
Cluster Labels: [np.int64(0), np.int64(0), np.int64(1), np.int64(2), np.int6
4(2), np.int64(3)]
Number of Clusters: 4
```

In [ ]:

```python

```